→ Q 11   35 → 37

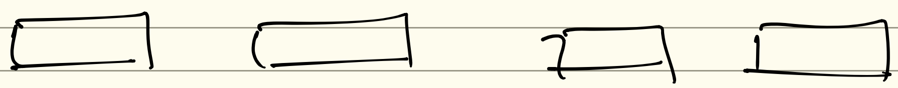
→ Selection
→ merge
→

→ 23   32   45   69   72   73   89   97

Sorted →

quick sort

Bubble / Inse-



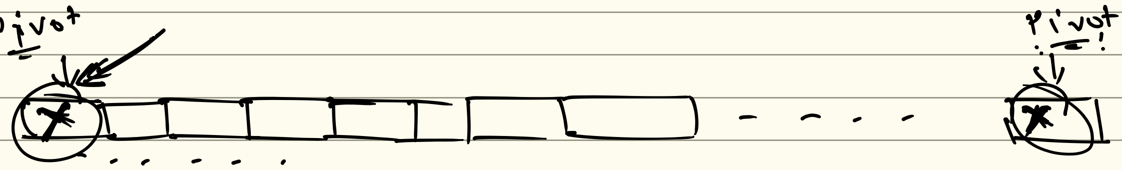
40 - 42 →

Ans →

Q 30 →   25 distinct → Quicksort - pivot

worst place ↔ probability.

$\frac{2}{25}$



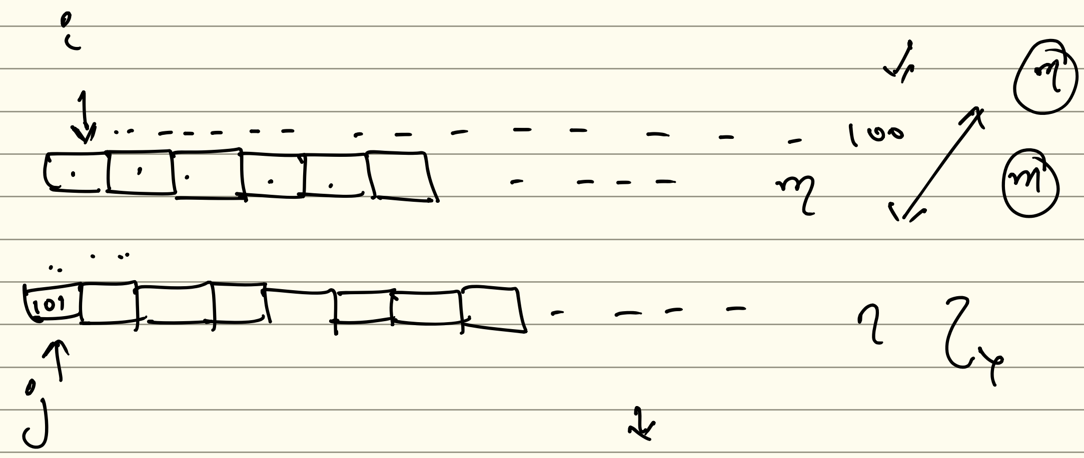
$\frac{2}{25} \Rightarrow 0.08$

Q5)

i)

48

q: 48

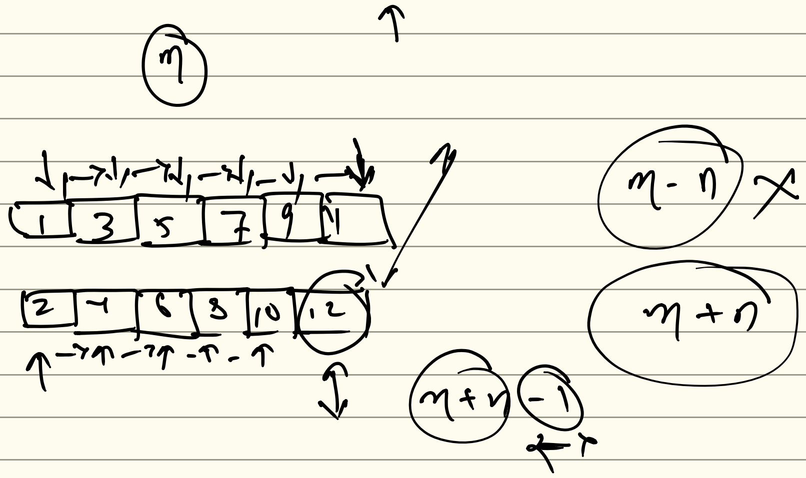


$m - n$

$\max(m, n)$

$\min(m, n)$

$m + n - 1$



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6)

52, 54

2   2

8   22   7   9   31   5   13   4

1^{st} pass   8   7   9   22   5   13   31 → 2

2^{nd} pass   7   8   9   5   13   22   31

7   8   5   9   13   22   31 → 1

7   5   8   9   13   22   31 → 1

5   7   8   9   13   22   31 → 1

a) 11 10 9 8 → $\frac{n(n-1)}{2}$

10:03

10:05

Quick

n^2

1 1

1 → $\theta \approx n$

→ $O(n^2)$

Q11) L=4

Two way straight

(1) 20 47 15 8 9 4 40 30 12 17

47 20 8 15 4 9 30 40 17 12

8 15 20 47 4 9 30 40 12 17

B

10:15 - 10:17

Q12 $\Rightarrow$ → start ?
$\Leftarrow$

100 sec ≈ 1000

n = 1000

$100 = 1000 \times \log 1000 \times c$ →

$c = \boxed{\dfrac{100}{1000 \times \log 1000}}$

$T = c \times 100 \times \log 100$

$\boxed{T = \frac{1}{10} \times \frac{1}{\log 1000} \times 100 \log 100}$ ⇐

$= \frac{1}{1\cancel{0} \times 3 \times \log 10} \times 10\cancel{0} \times 2 \times \cancel{\log 10}$

{ }, | T ⇒ $\frac{1}{3} \times 20 \Rightarrow 20/3 \Rightarrow 6.7$

log 1000

$\log 10^3$ 3 log 10 ⇐

Q1 - 2.3

1 — ..
2 — ..
4 — .
8 — .
16 — .
— .
32
—
6.4 ⇒ 6

→ 30 sec

$\boxed{30 = c \times 64 \log 64}$ ←

$T = c \times n \log n$ → worst case

$30 = c \times 64 \log 64$

$\left\{ c = \dfrac{30}{64 \times \log 64} \right\}$

$\Leftarrow$ $360 = c \times n \log n$

$\Leftarrow$ $360 = \frac{30}{64 \times \log 64} \times n \log n$

$\frac{\overset{12}{\cancel{360}} \times 64}{\cancel{30}} \times \log 64 = n \log n$

$3 \times 4 \times 8 \times 8 \times 6 = n \log n$

$3 \times 2 \times 2 \times 2^3 \times 2^3 \times 3 \times 2 = n \log n$

$\boxed{3^2 \times 2^8 \Rightarrow n \log n}$

$\log_2 64$

$\Rightarrow$ 2^9

$\boxed{9 \times 2^7 = \boxed{n \log n}}$ $\Leftarrow$

2^9 $\quad$ 2^9 $\quad$ 2^9 $\quad$ $\Rightarrow$

$\rightarrow$

$\boxed{10}$ $\quad$ 12 $\quad$ 9 $\quad$ 13 $\quad$ 14 $\quad$ 7

9 $\quad$ 7 $\qquad$ $\boxed{10}$ $\qquad$ 12 $\quad$ 13 $\quad$ 14

2

K = 3

k $\Rightarrow$ k^{th} smallest

10 12 9 13 14 7 6

partition

10

9 7 6 10 12 13 14

0 1 2 3 4 5 6

3 → 10 → 4th smallest

4th smallest

2nd smallest → 3

2nd

k = 2

6th smallest

4

3

9 7 10 12 15 13 14

4th smallest

2nd smallest

7 - 3 - 1

k=2 6

3

52 : 54

→ n logn ⇐

a b c

a b j

→ 1

2

→ ①

⇒ n logn

a b c

a b j

$n \to n$

n x   n logn ⇐

$n^2 \log n$

$\Theta(n^2)$ → worst case

Θ^2 →

n - - - -

n/4   3n/4 - - -

$\Theta(n^2)$

$$T(n) = T(n/4) + T(3n/4) + n \Rightarrow$$

→ $(n-1)/4$

$\rightarrow$ $T(n)$ - - - - n

$T(n/2)$ $T(n/2)$ $+ \, n/2 + n/2$

$n \log n$